UNIVERSITY
OF OSLO

Master's thesis

The title of my thesis

Any short subtitle

My Name

Mathematics
60 ECTS study points

Department of Mathematics
Faculty of Mathematics and Natural Sciences

Spring 2025

My Name

The title of my thesis

Any short subtitle

Supervisor:
The Name

Abstract

Here come 3–6 sentences describing your thesis.

This is a relatively short working example of the UiO template for a Master's thesis written in LaTeX. The example is based on Dag Langemyhr's excellent template compliant with the UiO designmanual from 2022, Martin Helsø's instructive examples (2017–2019) and Karoline Moe's seminars on writing in mathematical subjects (2014–2024). We show how some of the standard and most helpful features can be used efficiently, but please note that several of the chapters, sections, packages, tools and environments are optional to include in your thesis and that you must adapt and modify the template to suit your thesis and the traditions in your field of study. If in doubt, ask! Your supervisor and department have the final say in this.

Sammendrag

Her kommer Abstract på et annet språk. Det skal være et sammendrag på norsk i Ph.d.-avhandlinger. Det er like greit å ha det i en masteroppgave også. Det kan være utfordrende å skrive, men det er nyttig.

Dette er en relativt kort tekst der vi bruker UiOs mal for masteroppgaver...

Add new section about results in Chapter 1.

Contents

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Preface

The preface is a short and perhaps more personal description of the project, the purpose and your motivation for pursuing it. It should be short and sweet; and it is different from the acknowledgements.

This template has grown out of the work with seminars about scientificness and writing in mathematical subjects that was organized by the Department of Mathematics and the Science Library at the University of Oslo (UiO), and hosted by Karoline Moe, Ph.D., in the period from 2014 to 2024.

There has always been a tradition at the department that former students pass their LaTeX-template on to the next generation of students. However, this practice did not ensure compliance neither with standard traditions in scientific writing and reference management, nor with subsequent design manuals at UiO and typography. Moreover, some of the commands and packages in LaTeX used by students were outdated or in conflict, and there was clearly a need for more instruction.

This template tries to answer these issues. The main purpose of the template is to serve as a starting point for all students using LaTeX to write their Master's thesis. First of all it is a 'minimal' working example with the structure of a thesis. Secondly, We have included examples that most students frequently run into and that can take forever to figure out, even with all available tools, search engines and AI. Finally, we have tried to include a description or short text of the most essential genre-features or pit-falls for each chapter.

We sincerely hope that this template can be of some help in your work, and feel free to send the the LaTeX-group at the University of Oslo Library an e-mail on latexguru@ub.uio.no if you are stuck!

Acknowledgements

chapter:ack

The acknowledgements is where you thank everyone who have contributed to the thesis. It is mandatory to thank your supervisor, co-supervisor, your department and other parties that you have cooperated with; fellow students, a company etc.

The acknowledgements is usually written in a more personal tone than other parts of the thesis, and you should use the pronoun 'I' when you write. F.ex. I would like to thank Dag Langemyhr and Martin Helsø for their passion for LaTeX, their dedication to elegance and detail in typography, and for reading the package documentation at *The Comprehensive TeX Archive Network* (CTAN)[1] so that the rest of us can worry less about layout and more about The Millennium Prize Problems[2].

In the acknowledgements it is custom to include important events that have happened during the period where you were working on the project. Sometimes, this can be done with humor and creativity. But be extremely cautious! Humor changes over time, your life and perspective changes, and you never know who will read this part of your thesis. What seems funny now might be extremely insulting or embarrassing in a couple of years[3].

In fact, the acknowledgements is the part of the thesis that *everyone* who picks up your thesis will read, not just the examiner or the interested researcher. I recommend that you visualize your audience when you feel most creative: your supervisor, your parents, your grandmother, your ex, your future boss, your future co-workers, your future kids, and, perhaps most importantly, your future students.

The fact that your audience might be larger than you think, can not be stressed enough. And this applies to the entire thesis. Your thesis will eventually be deposited in and made openly available to the entire world via the institutional repository DUO. Because of the UiO-address, it gets a high rating on search engines and will be very easy to find for interested researchers all over the world. Moreover, most researchers subscribe to alerts when new citations of their articles or books are published, which means that most of the researchers that you have cited are notified about your thesis, and they might read it, push it to their students, find errors, notice sloppy work and referencing, pick up interesting results and more.

[1] Note that CTAN is the central place for all kinds of material around TeX, with 6649 packages as of October 2024.

[2] Footnotes are usually not used in mathematical texts, in particular not for citations! More on that later. In the meantime, use Oria via ub.uio.no to find books about The Millennium Prize Problems

[3] As a rule of thumb, try to refrain from being too creative and funny in your thesis. Write a blog, diary or send letters to a friend if you must.

Chapter 1

Results

results

1.1 Surfaces including intersecting lines

In this section we will study the problem of finding a smooth hypersurface $X_d \subset \mathbb{P}^3$ of degree d, containing k lines intersecting at a common point p and lying in a common plane H. We will denote the lines as $l_1, l_2, \ldots, l_k$.

Remark 1.1.1. Note that for automorphisms of $\mathbb{P}^3$, lines are sent to each other. Therefore we can study the lines $l_0 = (\alpha : \beta : 0 : 0)$, $l_1 = (\alpha : 0 : \gamma : 0)$, $l_3 = (\alpha : \delta : \lambda_3\delta : 0), \ldots, l_k = (\alpha : \delta : \lambda_k\delta : 0)$ for $\lambda_i \in \mathbb{N}$, take the automorphism of $\mathbb{P}^3$ which sends these lines to the given lines, and then study the hypersurface of degree d in $\mathbb{P}^3$ which contains these lines. This automorphism will also takes a smooth hypersurface of degree d to a smooth hypersurface of degree d, so showing the following properties for lines in this form is sufficient for the general case.

We begin with the following lemma of non-existence:

Lemma 1.1.2. *Given distinct lines $l_1, l_2, \ldots, l_k$ in $\mathbb{P}^3$ intersecting at a common point p and lying in a common plane H, there does not exist any smooth hypersurface of degree $2 \geq d \geq k-1$ such that the lines are contained in the hypersurface.*

Proof. Assume for contradiction that there exists a smooth hypersurface of degree $2 \geq d \geq k-1$ such that the lines are contained in the hypersurface. Then we can fix a line l_{k+1} in H, which does not intersect p, and which does not lie in the hypersurface X_d. l_{k+1} will intersect each of the lines at a point p_i, all being distinct since the lines only intersects in p, and thus no other point. l_{k+1} will also intersect the hypersurface at the points p_i, thus intersecting the hypersurface at k points. By Bezout's theorem, the intersection of a line with a hypersurface of degree d is at most d points, giving a contradiction. □

We now move on to the case of existence, with the following proposition:

Proposition 1.1.3. *For k lines in $\mathbb{P}^3$ intersecting at a common point p and lying in a common plane, we can find a smooth hypersurface of degree d such that the lines are contained in the hypersurface.*

Proof. Let $\mathfrak{L}$ be the linear system of hypersurfaces of degree d in $\mathbb{P}^3$ containing the lines $l_0, l_1, \ldots, l_k$. Bertini's theorem states that the general member of $\mathfrak{L}$ is smooth outside the base locus of $\mathfrak{L}$. The base locus of $\mathfrak{L}$ is the union of the lines $l_0, l_1, \ldots, l_k$. Thus we must find a function $f \in \mathbb{C}[x_0, x_1, x_2, x_3]_d$ such that $V(f)$ is smooth on the lines $l_0, l_1, \ldots, l_k$.

Bertini's theorem then gives us that we can find a hypersurface which is also smooth outside the base locus of $\mathfrak{L}$.

We study the function $f = g(x_1, x_2) + h(x_0, x_3)$, where $h = x_0^{k-1}x_3$ and $g = \prod l_i = x_1x_2\prod_{i=3}^{k}(x_1 - \lambda_i x_2)$, giving the following polynomial:

$$f(x_0, x_1, x_2, x_3) = x_1x_2\prod_{i=3}^{k}(x_1 - \lambda_i x_2) + x_0^{k-1}x_3$$

We can see that f is a polynomial of degree d, and that f includes the lines $l_0, l_1, \ldots, l_k$, since $f(\alpha : \beta : 0 : 0) = f(\alpha : 0 : \gamma : 0) = f(\alpha : \delta : \lambda_i\delta : 0) = 0$. We must then check that f is smooth on the base locus of $\mathfrak{L}$, which is to check that $\bigcap_{i=0}^{3}(\partial_i f = 0) = \emptyset$.

We get the following derivatives:

$$\begin{aligned}
\partial_0 f &= \partial_0 g + \partial_0 h = 0 + (d-1)x_0^{d-2}x_3 = (d-1)x_0^{d-2}x_3 \\
\partial_1 f &= \partial_1 g + \partial_1 h = x_2\prod_{i=3}^{k}(x_1 - \lambda_i x_2) + x_1x_2\sum_{i=3}^{k}\prod_{\substack{j=3\\ j\neq i}}^{k}(x_1 - \lambda_j x_2) \\
\partial_2 f &= \partial_2 g + \partial_2 h = x_1\prod_{i=3}^{k}(x_1 - \lambda_i x_2) + x_1x_2\sum_{i=3}^{k}(-\lambda_i)\cdot\prod_{\substack{j=3\\ j\neq i}}^{k}(x_1 - \lambda_j x_2) \\
\partial_3 f &= \partial_3 g + \partial_3 h = 0 + x_0^{d-1} = x_0^{d-1}
\end{aligned}$$

We then find that

- $\partial_0 f = 0$ on the entire base locus.
- $\partial_1 f = 0$ on the line $l_0 = (\alpha : \beta : 0 : 0)$.
- $\partial_2 f = 0$ on the line $l_1 = (\alpha : 0 : \gamma : 0)$.
- $\partial_3 f = 0$ if and only if $x_0 = 0$, which is the case for the following points: $(0 : \beta : 0 : 0), (0 : 0 : \gamma : 0), (0 : \delta : \lambda_i\delta : 0)$ for $3 \leq i \leq k$.

We can see that the intersection of all these zero locuses is empty, showing that the function f is smooth on the base locus of $\mathfrak{L}$. Thus we can find a smooth hypersurface of degree d such that the lines are contained in the hypersurface.

$\square$

By the results above, we get the following corollary:

Corollary 1.1.4. *Given k lines in $\mathbb{P}^3$ intersecting at a common point p and lying in a common plane, we can find a smooth hypersurface of degree d such that the lines are contained in the hypersurface if and only if $d \geq k$ or $d = 1$.*

Proof. If we have k lines, we can find a smooth hypersurface of degree $d \geq k + 1$ by including arbitrary lines $l_{k+1}, l_{k+2}, \ldots, l_d$. For the case $d = 1$, we can take the plane H containing the lines. $\square$